Linear Programming

Problem 1

A baker makes two types of treats: **Cupcakes (x)** and **Cookies (y)**.

- **Time Constraint:** It takes 2 hours to bake a batch of cupcakes and 1 hour for cookies. He has **8 hours** available.
- **Space Constraint:** He can only fit **5 total batches** in his display case.
- **Objective:** He makes a profit of **$30** per batch of cupcakes and **$20** per batch of cookies. Find the number of cupcake batches (x) and cookie batches (y) that maximize the profit.

Time constraint : (*) 1 cupcake takes 2 hours

x cupcake takes 2x hours.

(*) 1 cookies take 1 hour

y cookies take y hours

Total of hours to make x cupcakes and y cookies is $2x + y$.

since there is only 8 hours available

$$\boxed{2x + y \leq 8}$$

(*) Space constraint : $x + y \leq 5$

(*) we want to maximize : $30x + 20y$

(4) $x \geq 0$; $y \geq 0$

The Math: Problem 1 (Maximization)

Variables:

- x = Number of batches of Cupcakes
- y = Number of batches of Cookies

Objective Function:
Maximize the total profit (P):

$$P = 30x + 20y$$

Subject to Constraints:

1. **Time:** $2x + y \leq 8$ (Each cupcake batch takes 2 hours; cookies take 1)
2. **Space:** $x + y \leq 5$ (Total batches cannot exceed 5)
3. **Non-negativity:** $x \geq 0, y \geq 0$

Step - by - Step procedure

(1) List all the lines

(A) : $2x + y = 8$

(B) : $x + y = 5$

(C) : $x = 0$

(D) : $y = 0$

(2) Find <u>ALL possible intersections</u> of the lines

Intersection between (A), (B) :

$$\begin{array}{rl} & 2x+y=8 \\ - & x+y=5 \\ \hline & x \quad = 3 \end{array}$$

$\Rightarrow \quad y = 5 - x = 2 \quad \Rightarrow \quad y = 2$

Intersection is $(3, 2)$

(A) and (C) $\quad \begin{cases} 2x+y=8 \\ x=0 \end{cases} \Leftrightarrow \begin{cases} x=0 \\ y=8 \end{cases}$

(A) and (D) $\quad \begin{cases} 2x+y=8 \\ y=0 \end{cases} \Leftrightarrow \begin{cases} x=4 \\ y=0 \end{cases}$

(B) and (C) $\quad \begin{cases} x+y=5 \\ x=0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=5 \end{cases}$

(B) and (D) $\quad \begin{cases} x+y=5 \\ y=0 \end{cases} \Rightarrow \begin{cases} x=5 \\ y=0 \end{cases}$

(C) and (D): $\quad \begin{cases} x=0 \\ y=0 \end{cases}$

Step 3 : Find All the vertices.

All the vertices must satisfy the 2 constrain:

$$\begin{cases} 2x + y \leq 8 \\ x + y \leq 5 \end{cases}$$

x	y	Check for constraints
3	2	✓
0	8	X
4	0	✓
0	5	✓
5	0	X
0	0	✓

Step 5 : Check for max profit

x	y	30x + 20y
3	2	130
4	0	120
0	5	100
0	0	0

← maximum

He should make 3 batches of cupcakes and 2 batches of cookies to maximize the profit.

You try

A small shop builds two types of custom computers: Gaming PCs (x) and Office PCs (y).

- **Time Constraint**: A Gaming PC takes 3 hours to build, and an Office PC takes 1 hour. The shop has 9 hours of building time available today.
- **Component Constraint**: Due to a parts shortage, they can only build a total of 5 computers maximum.
- **Objective**: They earn $50 for every Gaming PC and $30 for every Office PC. Find the number of Gaming PCs (x) and Office PCs (y) that maximize the profit.

Problem 2

An athlete needs to meet daily nutritional requirements using **Protein Shakes (x)** and **Energy Bars (y)**.

- **Protein Constraint:** Each shake has 2 units of protein, and each bar has 1 unit. He needs at least **5 units** a day.
- **Energy Constraint:** Each shake has 1 unit of energy, and each bar has 3 units. He needs at least **10 units** a day.
- **Objective:** Each shake costs **$4** and each bar costs **$3**. Find the number of Protein Shakes (x) and Energy Bars (y) that minimize the cost.

(*) constraint

(1) $2x + y \geq 5$

(2) $x + 3y \geq 10$

(3) $x \geq 0$

(4) $y \geq 0$

Minimize cost $= 4x + 3y$

Step 1 : List all the lines

(A) $2x + y = 5$

(B) $x + 3y = 10$

(C) $x = 0$

(D) $y = 0$

Step 2 : Find All possible intersections between the lines

(A) and (B) :

$$\begin{cases} 2x + y = 5 \\ x + 3y = 10 \end{cases}$$

times 2 to both sides

$$\Rightarrow \quad - \begin{array}{r} 2x + y = 5 \\ 2x + 6y = 20 \\ \hline -5y = -15 \end{array}$$

$$\Rightarrow \quad \boxed{y = 3}$$

$$x + 3y = 10 \quad \Rightarrow \quad x = 10 - 3y$$

$$\Rightarrow \quad x = 10 - 3 \times 3$$

$$\Rightarrow \quad \boxed{x = 1}$$

$$\Rightarrow \quad (x, y) = (1, 3)$$

(A) and (C) :

$$\begin{cases} 2x + y = 5 \\ x = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 5 \end{cases}$$

(A) and (D) : $\begin{cases} 2x + y = 5 \\ y = 0 \end{cases} \Rightarrow \begin{cases} x = 2.5 \\ y = 0 \end{cases}$

(B) and (C) $\begin{cases} x + 3y = 10 \\ x = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 10/3 \end{cases}$

(B) and (D) $\begin{cases} x + 3y = 10 \\ y = 0 \end{cases} \Rightarrow \begin{cases} x = 10 \\ y = 0 \end{cases}$

(C) and (D) $\begin{cases} x = 0 \\ y = 0 \end{cases}$

<u>Step 3</u> : Check to find the vertices

x	y	$2x + y \geq 5$	$x + 3y \geq 10$	
1	3	✓	✓	Good
0	5	✓	✓	Good
2.5	0	✓	✗	Bad
0	10/3	✗	✓	Bad
10	0	✓	✓	Good
0	0	✗	✗	Bad

Step 4 : Check for min cost

x	y	cost = 4x + 3y
1	3	13
0	5	15
10	0	40

minimum

The athlete should buy 1 protein shake and 3 energy bar.

The Math: Problem 2 (Minimization)

Variables:

- x = Number of Protein Shakes
- y = Number of Energy Bars

Objective Function:
Minimize the total cost (C):

$$C = 4x + 3y$$

Subject to Constraints:

1. **Protein:** $2x + y \geq 5$
2. **Energy:** $x + 3y \geq 10$
3. **Non-negativity:** $x \geq 0, y \geq 0$

You try

A landscaping company needs to spread mulch and stone using two sizes of trucks: **Small Trucks (x)** and **Large Trucks (y)**.

- **Mulch Capacity Constraint:** A Small Truck carries 1 ton, and a Large Truck carries 3 tons. They need to deliver at least **6 tons** of mulch.
- **Stone Capacity Constraint:** A Small Truck carries 2 tons, and a Large Truck carries 1 ton. They need to deliver at least **7 tons** of stone.
- **Objective:** Running a Small Truck costs **$100** and a Large Truck costs **$200**. Find the number of Small Trucks (x) and Large Trucks (y) that minimize the cost.